KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

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MATHEMATICS

Year 9 5.2

Yearly Exam

2026

Marking Guide

Notes for markers

1. Full marks for a correct answer unless the question asks for working.
2. Follow-through: a later part that uses an earlier wrong answer correctly earns its marks.
3. Units: do not deduct for missing units; note it for feedback.
4. Rounding: deduct only where the criteria say so.
5. Green = Foundation, Blue = Standard (Test yourself levels).

Sections and Marks

Section 1 - Data — /25
17 green · 8 blue

Section 2 - Surface area and volume — /24
17 green · 7 blue

Section 3 - Mixed: Chapters 1-8 — /21
15 green · 6 blue

Total marks /70

Section	Marks	Share	Green	Blue	Questions
Section 1 - Data	25	35.7%	17	8	5 MC · 6 SA · 2 WP
Section 2 - Surface area and volume	24	34.3%	17	7	5 MC · 4 SA · 2 WP
Section 3 - Mixed: Chapters 1-8	21	30%	15	6	5 MC · 5 SA · 2 WP
Total	**70**	**100%**	**49 (70%)**	**21 (30%)**	

Section 1: Data – answer summary				
Q	**Answer**	**Marks**	**Level**	**Topic and source**
1	D (The time taken to swim 50 m)	1	Green	Classifying data TY9 Q9 (9.05, green)
2	B (Negatively skewed, clustered at 8 and 9)	1	Green	Shape of a distribution (dot plot) TY9 Q6a (9.03, green)
3	C (32)	1	Green	Median from a stem-and-leaf plot TY9 Q3b (9.02, green)
4	A (2)	1	Green	Mode from a histogram TY9 Q4b (9.02, green)
5	C (The mean)	1	Green	Effect of an outlier TY9 Q2c and Q3c-d (9.01, 9.02, green)
6	13 laps	1	Green	Mean of a data set TY9 Q1 (9.01, green)
7	12.5 laps	1	Green	Median of a data set TY9 Q1 (9.01, green)
8	12 laps	1	Green	Mode of a data set TY9 Q1 (9.01, green)
9	9 laps	1	Green	Range of a data set TY9 Q1 (9.01, green)
10a	$f \times x$: 0, 6, 16, 15, 12, 10; totals 26 and 59	1	Green	Frequency table: mean, median and range TY9 Q4 (9.02, green)
10b	2.27 visits	1	Green	Frequency table: mean, median and range TY9 Q4 (9.02, green)
10c	2 visits	1	Green	Frequency table: mean, median and range TY9 Q4 (9.02, green)
10d	5 visits	1	Green	Frequency table: mean, median and range TY9 Q4 (9.02, green)
11a	Stems 1 to 4, Pete's Pizza leaves on one side and Pizza Point leaves on the other, all in order. Key: 2 | 1 = 21 minutes.	2	Blue	Back-to-back stem-and-leaf plot; comparing two data sets TY9 Q5 (9.02, blue)
11b	Pete's Pizza: 26.5 min; Pizza Point: 20 min	1	Blue	Back-to-back stem-and-leaf plot; comparing two data sets TY9 Q5 (9.02, blue)
11c	Pizza Point: its median time (20 min) is less than the median for Pete's Pizza (26.5 min).	1	Blue	Back-to-back stem-and-leaf plot; comparing two data sets TY9 Q5 (9.02, blue)
12a	$1154	1	Green	Mean and median with an outlier; best measure of centre TY9 Q2 (9.01, green)
12b	$890	1	Green	Mean and median with an outlier; best measure of centre TY9 Q2 (9.01, green)
12c	$882.22	1	Green	Mean and median with an outlier; best measure of centre TY9 Q2 (9.01, green)

Section 1: Data – answer summary (continued)				
Q	**Answer**	**Marks**	**Level**	**Topic and source**
12d	The median ($890): the manager's wage of $3600 is an outlier that pulls the mean up to $1154, which is more than 9 of the 10 wages.	1	Green	Mean and median with an outlier; best measure of centre TY9 Q2 (9.01, green)
13a	Kestrels: 39.1; Tigers: 41.2	1	Blue	Comparing two data sets (back-to-back stem-and-leaf plot) TY9 Q7 (9.04, blue)
13b	Kestrels: 39; Tigers: 42.5	1	Blue	Comparing two data sets (back-to-back stem-and-leaf plot) TY9 Q7 (9.04, blue)
13c	The Tigers: their mean (41.2) and median (42.5) are both higher than the Kestrels' (39.1 and 39).	1	Blue	Comparing two data sets (back-to-back stem-and-leaf plot) TY9 Q7 (9.04, blue)
13d	The Kestrels: their range is 15 goals (46 – 31), much smaller than the Tigers' 35 goals (57 – 22).	1	Blue	Comparing two data sets (back-to-back stem-and-leaf plot) TY9 Q7 (9.04, blue)
	Total	**25**	17 green · 8 blue	

Section 2: Surface area and volume – answer summary				
Q	**Answer**	**Marks**	**Level**	**Topic and source**
1	C (4600 L)	1	Green	Metric units: capacity (kL to L) TY10 Q1i (10.01, green)
2	A (±25 mL)	1	Green	Precision and absolute error TY10 Q2 (10.02, green)
3	D (72 cm^2)	1	Green	Area of a parallelogram TY10 Q6a (10.05, green)
4	B (176 cm^2)	1	Green	Area of a kite TY10 Q6b (10.05, green)
5	C (450 cm^2)	1	Green	Area of a composite shape (frame) TY10 Q5b (10.04, green)
6	48 cm	2	Green	Perimeter of a composite shape (Pythagoras) TY10 Q4b (10.04, green)
7	168 cm^2	3	Green	Area of a composite shape (rectangle + triangle) TY10 Q5a (10.04, green)
8	4350 cm^2	2	Blue	Surface area of an open rectangular prism TY10 Q9b (10.07, blue)
9a	1696.5 cm^3	2	Blue	Volume and surface area of a cylinder TY10 Q12b and Q11b (10.09, 10.08, blue)
9b	791.7 cm^2	2	Blue	Volume and surface area of a cylinder TY10 Q12b and Q11b (10.09, 10.08, blue)
10a	60 m	1	Green	Perimeter and area of a composite shape (lawn) TY10 Q4a and Q5 (10.04, green)

Section 2: Surface area and volume – answer summary (continued)				
Q	**Answer**	**Marks**	**Level**	**Topic and source**
10b	166 m^2	2	Green	Perimeter and area of a composite shape (lawn) TY10 Q4a and Q5 (10.04, green)
10c	$2058.40	1	Green	Perimeter and area of a composite shape (lawn) TY10 Q4a and Q5 (10.04, green)
11a	2200 cm^2	2	Green	Prism volume and capacity (water trough) TY10 Q6d (green), Q15 and Q14 (blue), Q1i (green)
11b	330 000 cm^3	1	Blue	Prism volume and capacity (water trough) TY10 Q6d (green), Q15 and Q14 (blue), Q1i (green)
11c	330 L	1	Green	Prism volume and capacity (water trough) TY10 Q6d (green), Q15 and Q14 (blue), Q1i (green)
	Total	**24**	17 green · 7 blue	

Section 3: Mixed: Chapters 1-8 – answer summary				
Q	**Answer**	**Marks**	**Level**	**Topic and source**
1	C (47)	1	Green	Substitution TY1 Q2 (1.02, green)
2	B (10.9)	1	Green	Pythagoras' theorem: hypotenuse TY2 Q5 (2.03, green)
3	C ($123.25)	1	Green	Discount and selling price TY3 Q16 (3.06, green)
4	A ($12m^8$)	1	Green	Index laws: multiplying terms TY5 Q1 (5.01, green)
5	D (16)	1	Green	Networks: degree of a vertex TY6 Q5 (6.03, green)
6	$10x - 3$	2	Blue	Expanding and simplifying TY1 Q7 (1.07, blue)
7	$913.50	2	Green	Simple interest TY3 Q19 (3.07, green)
8	8.93 cm	2	Green	Trigonometry: finding a side TY4 Q10 (4.05, green)
9a	7.24×10^{-4}	1	Green	Scientific notation TY5 Q10 and Q13 (5.08, 5.09, green)
9b	1.68×10^5	1	Green	Scientific notation TY5 Q10 and Q13 (5.08, 5.09, green)
10a	$x = 7$	1	Green	Solving equations TY7 Q1 (7.01, green) and Q5 (7.03, blue)
10b	$m = 8$	1	Blue	Solving equations TY7 Q1 (7.01, green) and Q5 (7.03, blue)

Section 3: Mixed: Chapters 1-8 – answer summary (continued)				
Q	**Answer**	**Marks**	**Level**	**Topic and source**
11a	$1018.40	1	Green	Wages and overtime (award rates) TY8 Q2 (8.01, green) and Q4 (8.02, blue)
11b	$1299.80	2	Blue	Wages and overtime (award rates) TY8 Q2 (8.01, green) and Q4 (8.02, blue)
12a	4.95 m	1	Green	Ladder: Pythagoras and trigonometry TY2 Q8 (2.04, green), TY4 Q13 (4.07, green) and Q11 (4.06, blue)
12b	72°	1	Green	Ladder: Pythagoras and trigonometry TY2 Q8 (2.04, green), TY4 Q13 (4.07, green) and Q11 (4.06, blue)
12c	6.73 m	1	Blue	Ladder: Pythagoras and trigonometry TY2 Q8 (2.04, green), TY4 Q13 (4.07, green) and Q11 (4.06, blue)
	Total	**21**	15 green · 6 blue	

Section 1: Data – marking criteria

Question 1 1 mark Green Classifying data TY9 Q9 (9.05, green)

ANSWER **D (The time taken to swim 50 m)**
MARKS **1 mark:** D (The time taken to swim 50 m)
NOTES *Time is measured, so it is numerical continuous. A is numerical discrete (counted), B is categorical nominal, C is categorical ordinal.*

Question 2 1 mark Green Shape of a distribution (dot plot) TY9 Q6a (9.03, green)

ANSWER **B (Negatively skewed, clustered at 8 and 9)**
MARKS **1 mark:** B (Negatively skewed, clustered at 8 and 9)
NOTES *Most scores are high (a cluster at 8 and 9) with a long tail to the left (low scores), so the data is negatively skewed. (Mean 7.3 is below the median 8.)*

Question 3 1 mark Green Median from a stem-and-leaf plot TY9 Q3b (9.02, green)

ANSWER **C (32)**
MARKS **1 mark:** C (32)
NOTES *There are 15 ages, so the median is the 8th age: 32. (30 is the 7th age, 31 averages the 7th and 8th ages, 36 is the mode.)*

Question 4 1 mark Green Mode from a histogram TY9 Q4b (9.02, green)

ANSWER **A (2)**
MARKS **1 mark:** A (2)
NOTES *The tallest column is at 2 (frequency 9), so the mode is 2 people. 9 is the frequency, not the score.*

Question 5 1 mark Green Effect of an outlier TY9 Q2c and Q3c-d (9.01, 9.02, green)

ANSWER **C (The mean)**
MARKS **1 mark:** C (The mean)
NOTES *Mean 7.9 → 6.44 (changes by 1.46); median 6.5 → 6; mode 6 → 6 (no change).*

Question 6 1 mark Green Mean of a data set TY9 Q1 (9.01, green)

ANSWER **13 laps**
WORKING Sum = 9 + 15 + 11 + 12 + 18 + 12 + 14 + 10 + 16 + 13 = 130
Mean = 130 ÷ 10
MARKS **1 mark:** 13

Question 7 1 mark Green Median of a data set TY9 Q1 (9.01, green)

ANSWER **12.5 laps**
WORKING In order: 9, 10, 11, 12, 12, 13, 14, 15, 16, 18
10 scores: median = (12 + 13) ÷ 2
MARKS **1 mark:** 12.5
NOTES *Common error: 12 or 13 (one middle score only): 0 marks.*

Question 8 1 mark Green Mode of a data set TY9 Q1 (9.01, green)

ANSWER **12 laps**
WORKING 12 occurs 2 times; every other score occurs once.
MARKS **1 mark:** 12

Marking criteria (continued)

Question 9 1 mark ■ Green Range of a data set TY9 Q1 (9.01, green)

ANSWER **9 laps**
WORKING Range = 18 − 9
MARKS **1 mark:** 9

Question 10 4 marks ■ Green Frequency table: mean, median and range TY9 Q4 (9.02, green)

a) ANSWER ***f* × *x*: 0, 6, 16, 15, 12, 10; totals 26 and 59** 1 mark · ■ Green
MARKS **1 mark:** all f × x values and both totals (26 and 59) correct

b) ANSWER **2.27 visits** 1 mark · ■ Green
WORKING Mean = 59 ÷ 26 = 2.2692…
MARKS **1 mark:** 2.27
NOTES *Follow-through from part a. 2.3 (wrong rounding): 0 marks.*

c) ANSWER **2 visits** 1 mark · ■ Green
WORKING 26 scores: average of the 13th and 14th scores
Running totals 2, 8, 16, 21, 24, 26: both are 2
MARKS **1 mark:** 2
NOTES *Common error: 2.5 (the middle of the scores 0 to 5): 0 marks.*

d) ANSWER **5 visits** 1 mark · ■ Green
WORKING Range = 5 − 0
MARKS **1 mark:** 5
NOTES *Common error: 8 − 2 = 6 (uses the frequencies): 0 marks.*

Question 11 4 marks ● Blue Back-to-back stem-and-leaf plot; comparing two data sets TY9 Q5 (9.02, blue)

a) ANSWER **Stems 1 to 4, Pete's Pizza leaves on one side and Pizza Point leaves on the other, all in order. Key: 2 | 1 = 21 minutes.** 2 marks · ● Blue
WORKING Pete's Pizza | Stem | Pizza Point
8 | **1** | 5 6 7 8 9
9 7 6 5 4 1 | **2** | 1 2 3 8
5 2 | **3** | 0
1 | **4** | –
MARKS **1 mark:** stems 1 to 4 with all 10 leaves on one side correct and in order
1 mark: all 10 leaves on the other side correct and in order, and a key
NOTES *Either shop may be on the left. Leaves on the left may increase outwards from the stem (as shown) or be written smallest first; accept either if consistent. All leaves correct but not in order: 1 mark.*

b) ANSWER **Pete's Pizza: 26.5 min; Pizza Point: 20 min** 1 mark · ● Blue
WORKING Pete's Pizza: (26 + 27) ÷ 2
Pizza Point: (19 + 21) ÷ 2
MARKS **1 mark:** both medians correct
NOTES *Follow-through from the plot in part a.*

c) ANSWER **Pizza Point: its median time (20 min) is less than the median for Pete's Pizza (26.5 min).** 1 mark · ● Blue
MARKS **1 mark:** Pizza Point with a reason that compares the medians (or the means, or where the times cluster)
NOTES *Accept a comparison of means (Pizza Point 20.9 min, Pete's Pizza 27.8 min). Follow-through from part b.*

Marking criteria (continued)

Question 12 4 marks · Green · Mean and median with an outlier; best measure of centre · TY9 Q2 (9.01, green)

a) 1 mark · Green

ANSWER **$1154**

WORKING
Sum = $11 540
Mean = $11 540 ÷ 10

MARKS **1 mark:** $1154

b) 1 mark · Green

ANSWER **$890**

WORKING
In order: $760, $780, $840, $860, $880, $900, $920, $950, $1050, $3600
Median = ($880 + $900) ÷ 2

MARKS **1 mark:** $890

c) 1 mark · Green

ANSWER **$882.22**

WORKING
Sum = $11 540 − $3600 = $7940
Mean = $7940 ÷ 9 = $882.2222…

MARKS **1 mark:** $882.22 (correct to the nearest cent)

d) 1 mark · Green

ANSWER **The median ($890): the manager's wage of $3600 is an outlier that pulls the mean up to $1154, which is more than 9 of the 10 wages.**

MARKS **1 mark:** median, with a reason about the outlier (the manager's wage) making the mean too high

NOTES *Part c supports this: without the manager the mean is $882.22, close to the median.*

Question 13 4 marks · Blue · Comparing two data sets (back-to-back stem-and-leaf plot) · TY9 Q7 (9.04, blue)

a) 1 mark · Blue

ANSWER **Kestrels: 39.1; Tigers: 41.2**

WORKING
Kestrels: 469 ÷ 12 = 39.083…
Tigers: 494 ÷ 12 = 41.167…

MARKS **1 mark:** both means correct to 1 decimal place

b) 1 mark · Blue

ANSWER **Kestrels: 39; Tigers: 42.5**

WORKING
Kestrels: (38 + 40) ÷ 2
Tigers: (41 + 44) ÷ 2

MARKS **1 mark:** both medians correct

c) 1 mark · Blue

ANSWER **The Tigers: their mean (41.2) and median (42.5) are both higher than the Kestrels' (39.1 and 39).**

MARKS **1 mark:** Tigers, with a reason that compares the means or the medians

NOTES *Follow-through from parts a and b.*

d) 1 mark · Blue

ANSWER **The Kestrels: their range is 15 goals (46 − 31), much smaller than the Tigers' 35 goals (57 − 22).**

MARKS **1 mark:** Kestrels, with both ranges (15 and 35) used to explain

Section 2: Surface area and volume – marking criteria

Question 1 1 mark · Green · Metric units: capacity (kL to L) · TY10 Q1i (10.01, green)

ANSWER **C (4600 L)**

MARKS **1 mark:** C (4600 L)

NOTES *1 kL = 1000 L, so 4.6 × 1000 = 4600 L. The others use × 100, × 10 000 or ÷ 1000.*

Marking criteria (continued)

Question 2 1 mark Green Precision and absolute error TY10 Q2 (10.02, green)

ANSWER **A (±25 mL)**

MARKS **1 mark:** A (±25 mL)

NOTES *The marks are 50 mL apart, so the precision is 50 mL and the absolute error is ½ × 50 = ±25 mL. B is the precision, C the gap between the numbered marks, D half of 1 mL.*

Question 3 1 mark Green Area of a parallelogram TY10 Q6a (10.05, green)

ANSWER **D (72 cm²)**

MARKS **1 mark:** D (72 cm²)

NOTES *A = 12 × 6 = 72 cm² (the height is the perpendicular height, 6 cm, not the sloping side).*

Question 4 1 mark Green Area of a kite TY10 Q6b (10.05, green)

ANSWER **B (176 cm²)**

MARKS **1 mark:** B (176 cm²)

NOTES *A = ½ × 16 × 22 = 176 cm². 352 cm² forgets the ½.*

Question 5 1 mark Green Area of a composite shape (frame) TY10 Q5b (10.04, green)

ANSWER **C (450 cm²)**

MARKS **1 mark:** C (450 cm²)

NOTES *30 × 25 − 20 × 15 = 750 − 300 = 450 cm².*

Question 6 2 marks Green Perimeter of a composite shape (Pythagoras) TY10 Q4b (10.04, green)

ANSWER **48 cm**

WORKING Sloping side: across 12 cm, up 14 − 9 = 5 cm
$c^2 = 12^2 + 5^2 = 169$, so $c = \sqrt{169} = 13$ cm
$P = 12 + 14 + 13 + 9$

MARKS **1 mark:** sloping side 13 cm (Pythagoras with 12 cm and 5 cm)
1 mark: perimeter 48 cm

NOTES *Follow-through: a wrong sloping side added correctly to the other three sides earns the second mark.*

Question 7 3 marks Green Area of a composite shape (rectangle + triangle) TY10 Q5a (10.04, green)

ANSWER **168 cm²**

WORKING Rectangle: 14 × 9 = 126 cm²
Triangle: ½ × 14 × 6 = 42 cm²
Total: 126 + 42

MARKS **1 mark:** rectangle 126 cm²
1 mark: triangle 42 cm²
1 mark: total area 168 cm²

NOTES *Follow-through: the third mark for correctly adding the student's two areas. Forgetting the ½ in the triangle (84 cm², total 210 cm²) loses the triangle mark only.*

Marking criteria (continued)

Question 8 2 marks ● Blue Surface area of an open rectangular prism TY10 Q9b (10.07, blue)

ANSWER **4350 cm²**

WORKING
Base: 45 × 30 = 1350 cm² (no top)
Front and back: 2 × 45 × 20 = 1800 cm²
Two ends: 2 × 30 × 20 = 1200 cm²
SA = 1350 + 1800 + 1200

MARKS **1 mark:** correct face areas: base 1350 cm², front and back 900 cm² each, ends 600 cm² each
1 mark: surface area 4350 cm² (five faces: no top)

NOTES *A closed-box answer of 5700 cm² (six faces) earns the first mark only.*

Question 9 4 marks ● Blue Volume and surface area of a cylinder TY10 Q12b and Q11b (10.09, 10.08, blue)

a) 2 marks · ● Blue

ANSWER **1696.5 cm³**

WORKING
$r = 12 \div 2 = 6$ cm
$V = \pi \times 6^2 \times 15 = 1696.460\ldots$

MARKS **1 mark:** correct substitution into $V = \pi r^2 h$: $\pi \times 6^2 \times 15$
1 mark: 1696.5 cm³ (needs $r = 6$)

NOTES *Using the diameter as the radius ($\pi \times 12^2 \times 15 = 6785.8$ cm³) earns the first mark only.*

b) 2 marks · ● Blue

ANSWER **791.7 cm²**

WORKING
$SA = 2 \times \pi \times 6^2 + 2 \times \pi \times 6 \times 15$
$= 226.19\ldots + 565.49\ldots$

MARKS **1 mark:** correct area of the two circular ends ($2 \times \pi \times 6^2 = 226.2$ cm²) or of the curved surface ($2 \times \pi \times 6 \times 15 = 565.5$ cm²)
1 mark: 791.7 cm²

NOTES *Follow-through from the radius used in part a. Leaving out one or both circular ends (678.6 or 565.5 cm²) earns the first mark only.*

Question 10 4 marks ■ Green Perimeter and area of a composite shape (lawn) TY10 Q4a and Q5 (10.04, green)

a) 1 mark · ■ Green

ANSWER **60 m**

WORKING
Missing sides: 18 − 8 = 10 m and 12 − 7 = 5 m
P = 18 + 7 + 10 + 5 + 8 + 12

MARKS **1 mark:** 60 m

b) 2 marks · ■ Green

ANSWER **166 m²**

WORKING
Bottom rectangle: 18 × 7 = 126 m²
Top rectangle: 8 × 5 = 40 m²
Area = 126 + 40

MARKS **1 mark:** splits the lawn into two rectangles with correct areas (126 m² and 40 m², or 216 m² − 50 m²)
1 mark: 166 m²

c) 1 mark · ■ Green

ANSWER **$2058.40**

WORKING 166 × $12.40

MARKS **1 mark:** $2058.40

NOTES *Follow-through from part b.*

Marking criteria (continued)

Question 11 | 4 marks | Green + Blue | Prism volume and capacity (water trough) | TY10 Q6d (green), Q15 and Q14 (blue), Q1i (green)

a)
ANSWER **2200 cm²** — 2 marks · Green
WORKING $A = ½ \times 40 \times (64 + 46)$
$= ½ \times 40 \times 110$
MARKS **1 mark:** correct substitution: $½ \times 40 \times (64 + 46)$
1 mark: 2200 cm²

b)
ANSWER **330 000 cm³** — 1 mark · Blue
WORKING $V = 2200 \times 150$
MARKS **1 mark:** 330 000 cm³
NOTES *Follow-through from part a.*

c)
ANSWER **330 L** — 1 mark · Green
WORKING $330\,000 \div 1000$
MARKS **1 mark:** 330 L
NOTES *Follow-through from part b.*

Section 3: Mixed: Chapters 1-8 – marking criteria

Question 1 | 1 mark | Green | Substitution | TY1 Q2 (1.02, green)

ANSWER **C (47)**
MARKS **1 mark:** C (47)
NOTES *$2a^2 - b = 2 \times 5^2 - 3 = 2 \times 25 - 3 = 47$. (97 squares 2a: $(2 \times 5)^2 - 3$; 17 treats a^2 as 2a; 7 leaves out the square.)*

Question 2 | 1 mark | Green | Pythagoras' theorem: hypotenuse | TY2 Q5 (2.03, green)

ANSWER **B (10.9)**
MARKS **1 mark:** B (10.9)
NOTES *$x^2 = 5.6^2 + 9.3^2 = 31.36 + 86.49 = 117.85$, so $x = \sqrt{117.85} = 10.856... \approx 10.9$. (14.9 adds the sides, 7.4 subtracts the squares, 3.7 subtracts the sides.)*

Question 3 | 1 mark | Green | Discount and selling price | TY3 Q16 (3.06, green)

ANSWER **C ($123.25)**
MARKS **1 mark:** C ($123.25)
NOTES *Discount = 15% × $145 = $21.75; selling price = $145 − $21.75 = $123.25. ($21.75 is the discount only, $130.00 takes off $15, $166.75 increases the price.)*

Question 4 | 1 mark | Green | Index laws: multiplying terms | TY5 Q1 (5.01, green)

ANSWER **A ($12m^8$)**
MARKS **1 mark:** A ($12m^8$)
NOTES *Multiply the numbers and add the indices: $4 \times 3 = 12$ and $m^6 \times m^2 = m^8$. ($7m^8$ adds the numbers, $12m^{12}$ multiplies the indices, $12m^4$ subtracts the indices.)*

Marking criteria (continued)

Question 5 1 mark ■ Green Networks: degree of a vertex TY6 Q5 (6.03, green)

ANSWER **D (16)**

MARKS **1 mark:** D (16)

NOTES *Degrees: P = 2, Q = 3, R = 4, S = 4, T = 2, U = 1; sum = 2 + 3 + 4 + 4 + 2 + 1 = 16 (= 2 × 8 edges). (6 is the number of vertices, 8 the number of edges, 12 is 2 × the number of vertices.)*
TY6 Q5 sits in a PATH section of the textbook (6.03). It is used because the brief asks for Chapter 6 networks only.

Question 6 2 marks ● Blue Expanding and simplifying TY1 Q7 (1.07, blue)

ANSWER **$10x - 3$**

WORKING $= 12x - 15 - 2x + 12$

MARKS **1 mark:** correct expansion of both brackets: $12x - 15 - 2x + 12$
1 mark: $10x - 3$

NOTES *Common error: $10x - 27$ ($-2 \times (-6)$ taken as -12): 1 mark.*

Question 7 2 marks ■ Green Simple interest TY3 Q19 (3.07, green)

ANSWER **$913.50**

WORKING $I = \$7250 \times 0.042 \times 3$

MARKS **1 mark:** correct substitution: $\$7250 \times 0.042 \times 3$
1 mark: $913.50

NOTES *Common error: $r = 4.2$ instead of 0.042 ($91 350.00): 1 mark.*

Question 8 2 marks ■ Green Trigonometry: finding a side TY4 Q10 (4.05, green)

ANSWER **8.93 cm**

WORKING $\sin 38° = \frac{x}{14.5}$
$x = 14.5 \times \sin 38° = 8.9271\ldots$

MARKS **1 mark:** $\sin 38° = x \div 14.5$ (or $x = 14.5 \sin 38°$)
1 mark: 8.93 cm

NOTES *Common errors: $14.5 \times \cos 38° = 11.43$ or $14.5 \div \sin 38° = 23.55$: 0 marks.*

Question 9 2 marks ■ Green Scientific notation TY5 Q10 and Q13 (5.08, 5.09, green)

a) 1 mark · ■ Green

ANSWER **7.24×10^{-4}**

WORKING Move the decimal point 4 places to the right.

MARKS **1 mark:** 7.24×10^{-4}

b) 1 mark · ■ Green

ANSWER **1.68×10^{5}**

WORKING $= 4.8 \times 3.5 \times 10^{6} \times 10^{-2}$
$= 16.8 \times 10^{4}$

MARKS **1 mark:** 1.68×10^{5}

NOTES *16.8×10^{4} is not in scientific notation: 0 marks. Accept 168 000 only if also written as 1.68×10^{5}.*

Marking criteria (continued)

Question 10	2 marks	Green + Blue	Solving equations	TY7 Q1 (7.01, green) and Q5 (7.03, blue)

a)

ANSWER: ***x* = 7** — 1 mark · Green

WORKING: $5x = 35$

MARKS: **1 mark:** $x = 7$

b)

ANSWER: ***m* = 8** — 1 mark · Blue

WORKING:
$7m - 2m - 6 = 3m + 10$
$5m - 6 = 3m + 10$
$2m = 16$

MARKS: **1 mark:** $m = 8$

NOTES: *Common error: −2 × 3 taken as +6, giving m = 2: 0 marks.*

Question 11	3 marks	Green + Blue	Wages and overtime (award rates)	TY8 Q2 (8.01, green) and Q4 (8.02, blue)

a)

ANSWER: **$1018.40** — 1 mark · Green

WORKING: 38 × $26.80

MARKS: **1 mark:** $1018.40

b)

ANSWER: **$1299.80** — 2 marks · Blue

WORKING:
Time-and-a-half: 3 × 1.5 × $26.80 = $120.60
Double time: 3 × 2 × $26.80 = $160.80
Total: $1018.40 + $120.60 + $160.80

MARKS: **1 mark:** overtime pay at both rates: $120.60 and $160.80 (or $281.40 in total)
1 mark: total $1299.80

NOTES: *Follow-through from part a.*
Common error: all 6 extra hours at time-and-a-half ($1259.60): 1 mark.

Question 12	3 marks	Green + Blue	Ladder: Pythagoras and trigonometry	TY2 Q8 (2.04, green), TY4 Q13 (4.07, green) and Q11 (4.06, blue)

a)

ANSWER: **4.95 m** — 1 mark · Green

WORKING:
$h^2 = 5.2^2 - 1.6^2 = 27.04 - 2.56 = 24.48$
$h = \sqrt{24.48} = 4.9477\ldots$

MARKS: **1 mark:** 4.95 m

NOTES: *Common error: adding the squares, $\sqrt{(5.2^2 + 1.6^2)} = 5.44$: 0 marks.*

b)

ANSWER: **72°** — 1 mark · Green

WORKING:
$\cos\theta = \frac{1.6}{5.2}$
$\theta = \cos^{-1}(1.6 \div 5.2) = 72.08\ldots°$

MARKS: **1 mark:** 72°

NOTES: *Accept sin or tan with the answer to part a (follow-through): both give the same angle to the nearest degree.*

c)

ANSWER: **6.73 m** — 1 mark · Blue

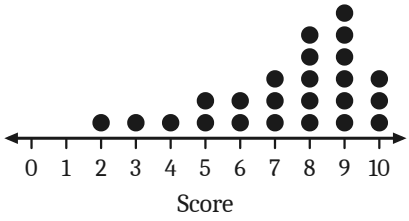
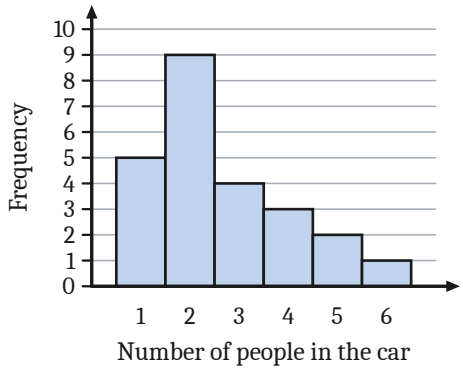
WORKING:
$\sin 75° = \frac{6.5}{L}$
$L = 6.5 \div \sin 75° = 6.7293\ldots$

MARKS: **1 mark:** 6.73 m

NOTES: *Common error: 6.5 × sin 75° = 6.28 m (unknown treated as in the numerator): 0 marks.*

Marking guide: answers in red

Section 1: Data

	Multiple Choice:	Mark
1	Which of the following is **numerical continuous** data? A. The number of goals in a hockey match B. The colours of cars in a car park C. Shirt sizes (S, M, L, XL) **D. The time taken to swim 50 m**	1
2	The dot plot shows the scores (out of 10) of 24 students in a spelling quiz. Which statement best describes the data? A. Positively skewed, clustered at 8 and 9 **B. Negatively skewed, clustered at 8 and 9** C. Negatively skewed, clustered at 2 and 3 D. Symmetrical, clustered at 8 and 9 	1
3	The stem-and-leaf plot shows the ages of the 15 people in a yoga class. What is the median age? A. 30 B. 31 **C. 32** D. 36 Stem | Leaf 1 | 8 9 2 | 3 5 5 7 3 | 0 2 4 6 6 6 4 | 1 5 5 | 8 Key: 1 | 8 = 18 years	1
4	The histogram shows the number of people in each of 24 cars that arrived at a beach car park. What is the mode? **A. 2** B. 3.5 C. 6 D. 9 	1
5	A water polo team scored these numbers of goals in 10 games: 5, 7, 6, 8, 6, 21, 7, 5, 6, 8 Which measure of centre changes the most if the outlier, 21, is removed? A. The mode B. The median **C. The mean** D. All three change by the same amount	1

Section 1: Data (continued)

Short Answer:

	Question	Working Out	Mark
	Use this set of data for questions 6 to 9. The numbers of laps swum by 10 students at swimming training were: **9, 15, 11, 12, 18, 12, 14, 10, 16, 13**		
6	Find the mean. Note: Mean = sum of the scores ÷ number of scores	Sum = 9 + 15 + 11 + 12 + 18 + 12 + 14 + 10 + 16 + 13 = 130 Mean = 130 ÷ 10 **13 laps**	1
7	Find the median.	In order: 9, 10, 11, 12, 12, 13, 14, 15, 16, 18 10 scores: median = (12 + 13) ÷ 2 **12.5 laps**	1
8	Find the mode.	12 occurs 2 times; every other score occurs once. **12 laps**	1
9	Find the range. Note: Range = highest score − lowest score	Range = 18 − 9 **9 laps**	1
10	A Year 9 class was asked how many times each student went to the beach last week. **Visits (x) / Frequency (f) / $f \times x$** 0 / 2 / **0** 1 / 6 / **6** 2 / 8 / **16** 3 / 5 / **15** 4 / 3 / **12** 5 / 2 / **10** **Total** / **26** / **59** a. Complete the table. b. Find the mean number of visits, correct to 2 decimal places. c. Find the median number of visits. d. Find the range. Note: Mean = total of $f \times x$ ÷ total frequency	a. **$f \times x$: 0, 6, 16, 15, 12, 10; totals 26 and 59** b. Mean = 59 ÷ 26 = 2.2692… **2.27 visits** c. 26 scores: average of the 13th and 14th scores Running totals 2, 8, 16, 21, 24, 26: both are 2 **2 visits** d. Range = 5 − 0 **5 visits**	1 1 1 1

Marking guide: answers in red

Section 1: Data (continued)

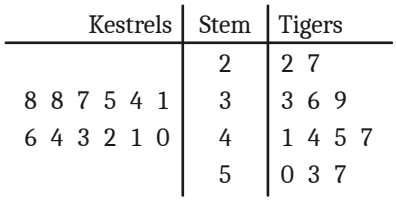
	Question	Working Out	Mark
11	The delivery times (in minutes) of 10 orders from each of two pizza shops are shown below. **Pete's Pizza**: 18, 25, 32, 21, 27, 35, 29, 24, 41, 26 **Pizza Point**: 15, 22, 19, 28, 17, 23, 16, 30, 21, 18		
a	Draw an ordered back-to-back stem-and-leaf plot of the data. Include a key.	Pete's Pizza | Stem | Pizza Point 8 | **1** | 5 6 7 8 9 9 7 6 5 4 1 | **2** | 1 2 3 8 5 2 | **3** | 0 1 | **4** | – **Stems 1 to 4, Pete's Pizza leaves on one side and Pizza Point leaves on the other, all in order. Key: 2 | 1 = 21 minutes.**	2
b	Find the median delivery time for each shop.	Pete's Pizza: (26 + 27) ÷ 2 Pizza Point: (19 + 21) ÷ 2 **Pete's Pizza: 26.5 min; Pizza Point: 20 min**	1
c	Which shop delivers faster? Give a reason.	**Pizza Point: its median time (20 min) is less than the median for Pete's Pizza (26.5 min).**	1

Marking guide: answers in red

	Section 1: Data (continued)		
	Worded Problems:		
	Question	**Working Out**	**Mark**
12	A surf shop has 10 employees, including the manager. Their weekly wages are: $860, $920, $780, $1050, $840, $3600, $900, $880, $760, $950 Note: Mean = sum of the scores ÷ number of scores		
a	Find the mean weekly wage.	Sum = $11 540 Mean = $11 540 ÷ 10 **$1154**	1
b	Find the median weekly wage.	In order: $760, $780, $840, $860, $880, $900, $920, $950, $1050, $3600 Median = ($880 + $900) ÷ 2 **$890**	1
c	The manager earns $3600. Find the mean weekly wage of the other 9 employees, correct to the nearest cent.	Sum = $11 540 − $3600 = $7940 Mean = $7940 ÷ 9 = $882.2222… **$882.22**	1
d	Which measure, the mean or the median, better describes the typical weekly wage at this shop? Give a reason.	**The median ($890): the manager's wage of $3600 is an outlier that pulls the mean up to $1154, which is more than 9 of the 10 wages.**	1

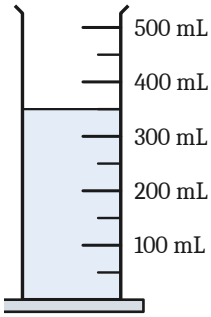
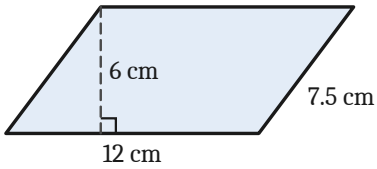
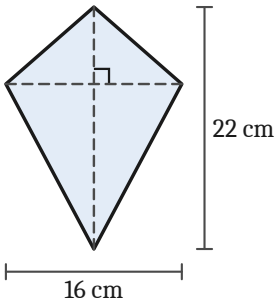
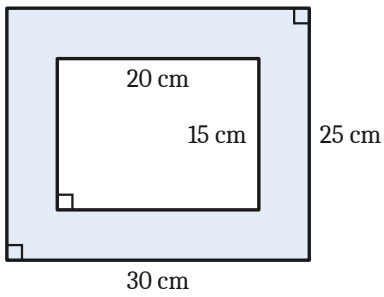
Marking guide: answers in red

Section 1: Data (continued)

	Question	Working Out	Mark
13	The Kestrels and the Tigers are two netball teams. The back-to-back stem-and-leaf plot shows the number of goals each team scored in its 12 games this season. Note: Mean = sum of the scores ÷ number of scores Note: Range = highest score − lowest score	 Key: 4 | 1 = 41 goals	
a	Find the mean number of goals for each team, correct to 1 decimal place.	Kestrels: 469 ÷ 12 = 39.083… Tigers: 494 ÷ 12 = 41.167… **Kestrels: 39.1; Tigers: 41.2**	1
b	Find the median number of goals for each team.	Kestrels: (38 + 40) ÷ 2 Tigers: (41 + 44) ÷ 2 **Kestrels: 39; Tigers: 42.5**	1
c	Which team usually scores more goals? Give a reason.	**The Tigers: their mean (41.2) and median (42.5) are both higher than the Kestrels' (39.1 and 39).**	1
d	Which team is more consistent? Use the range of each team to explain.	**The Kestrels: their range is 15 goals (46 − 31), much smaller than the Tigers' 35 goals (57 − 22).**	1
	Total		**/25**

Marking guide: answers in red

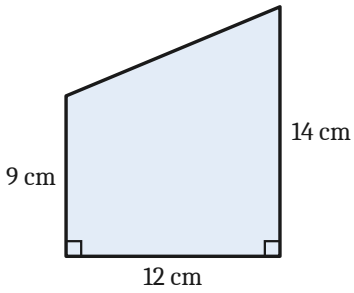
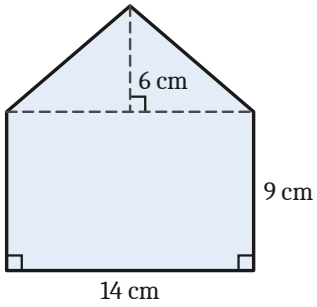
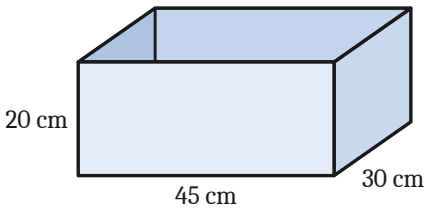
Section 2: Surface area and volume

	Multiple Choice:	Mark
1	A rainwater tank holds 4.6 kL of water. How many litres is this? A. 460 L B. 46 000 L **C. 4600 L** D. 0.0046 L	1
2	The scale on this measuring cylinder is shown. What is the absolute error of a reading taken with this measuring cylinder? Note: absolute error = ½ × precision (the smallest marked unit on the scale) **A. ±25 mL** B. ±50 mL C. ±100 mL D. ±0.5 mL 	1
3	What is the area of this parallelogram? Note: Area of a parallelogram: $A = bh$ A. 90 cm^2 B. 36 cm^2 C. 39 cm^2 **D. 72 cm^2** 	1
4	What is the area of this kite? Note: Area of a kite: $A = ½xy$ (x and y are the diagonals) A. 352 cm^2 **B. 176 cm^2** C. 88 cm^2 D. 38 cm^2 	1
5	A photo frame is a 30 cm by 25 cm rectangle with a 20 cm by 15 cm hole for the photo. What is the area of the frame (the shaded part)? Note: Area of a rectangle: $A = lb$ A. 750 cm^2 B. 1050 cm^2 **C. 450 cm^2** D. 100 cm^2 	1

Marking guide: answers in red

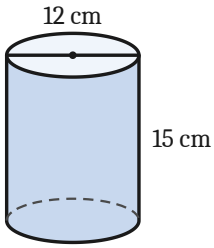
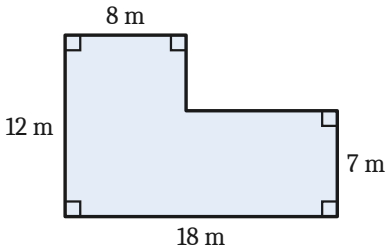
Section 2: Surface area and volume (continued)

Short Answer:

	Question	Working Out	Mark
6	Find the perimeter of this shape.  Note: Pythagoras' theorem: $c^2 = a^2 + b^2$	Sloping side: across 12 cm, up 14 − 9 = 5 cm $c^2 = 12^2 + 5^2 = 169$, so $c = \sqrt{169} = 13$ cm $P = 12 + 14 + 13 + 9$ **48 cm**	2
7	Find the area of this shape.  Note: Area of a rectangle: $A = lb$ Note: Area of a triangle: $A = \frac{1}{2}bh$	Rectangle: $14 \times 9 = 126$ cm^2 Triangle: $\frac{1}{2} \times 14 \times 6 = 42$ cm^2 Total: 126 + 42 **168 cm^2**	3
8	A storage tub is a rectangular prism with **no lid** (it is open at the top). Find the surface area of the tub.  Note: Surface area = the total area of the faces; area of a rectangle: $A = lb$	Base: $45 \times 30 = 1350$ cm^2 (no top) Front and back: $2 \times 45 \times 20 = 1800$ cm^2 Two ends: $2 \times 30 \times 20 = 1200$ cm^2 $SA = 1350 + 1800 + 1200$ **4350 cm^2**	2

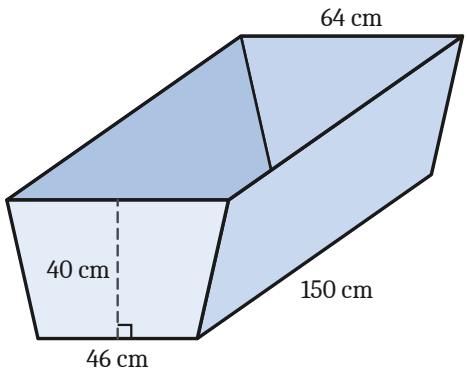
Marking guide: answers in red

Section 2: Surface area and volume (continued)

	Question	Working Out	Mark
9	This closed cylinder has a diameter of 12 cm and a height of 15 cm.  a. Find its volume, correct to 1 decimal place. b. Find its surface area, correct to 1 decimal place. Note: Volume: $V = \pi r^2 h$ Note: Surface area: $SA = 2\pi r^2 + 2\pi rh$	a. $r = 12 \div 2 = 6$ cm $V = \pi \times 6^2 \times 15 = 1696.460\ldots$ **1696.5 cm³** b. $SA = 2 \times \pi \times 6^2 + 2 \times \pi \times 6 \times 15$ $= 226.19\ldots + 565.49\ldots$ **791.7 cm²**	2 2
	Worded Problems:		
	Question	**Working Out**	**Mark**
10	The plan shows an L-shaped lawn at a primary school. All of the corners are right angles. Note: Area of a rectangle: $A = lb$  		
a	Garden edging is laid around the outside of the lawn. How many metres of edging are needed?	Missing sides: 18 − 8 = 10 m and 12 − 7 = 5 m $P = 18 + 7 + 10 + 5 + 8 + 12$ **60 m**	1
b	Find the area of the lawn.	Bottom rectangle: $18 \times 7 = 126$ m² Top rectangle: $8 \times 5 = 40$ m² Area = 126 + 40 **166 m²**	2
c	New turf costs $12.40 per square metre. Find the cost of the turf for the whole lawn.	166 × $12.40 **$2058.40**	1

Marking guide: answers in red

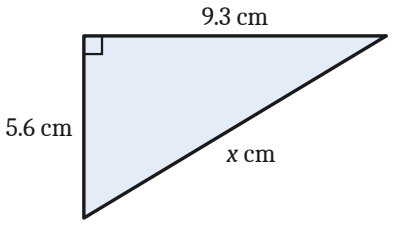
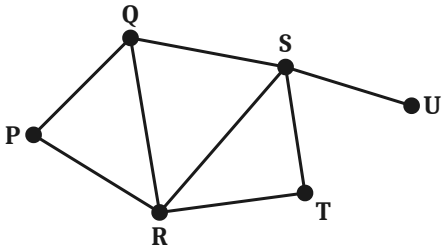
Section 2: Surface area and volume (continued)

	Question	Working Out	Mark
11	A farmer's water trough is a prism 150 cm long, open at the top. Its end is a trapezium with the measurements shown. 		
a	Find the area of the trapezium-shaped end of the trough. Note: Trapezium: $A = \frac{1}{2}h(a + b)$	$A = \frac{1}{2} \times 40 \times (64 + 46)$ $= \frac{1}{2} \times 40 \times 110$ **2200 cm²**	2
b	Find the volume of the trough in cm³. Note: Volume of a prism: $V = Ah$ (A = area of the end, h = length of the prism)	$V = 2200 \times 150$ **330 000 cm³**	1
c	How many litres of water does the trough hold when it is full? Note: 1 L = 1000 cm³	$330\,000 \div 1000$ **330 L**	1
	Total		**/24**

Marking guide: answers in red

Section 3: Mixed: Chapters 1-8

Multiple Choice:

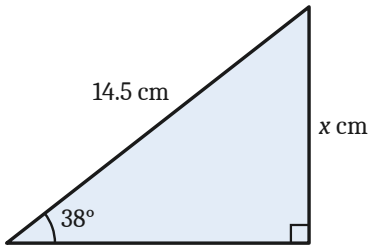
		Mark
1	If $a = 5$ and $b = 3$, evaluate $2a^2 - b$. A. 7 B. 17 **C. 47** D. 97	1
2	Find the value of x. Select the closest answer. Note: Pythagoras' theorem: $c^2 = a^2 + b^2$ A. 14.9 **B. 10.9** C. 7.4 D. 3.7 	1
3	A skateboard is marked at \$145. In a sale it is discounted by 15%. What is the selling price? A. \$166.75 B. \$130.00 **C. \$123.25** D. \$21.75	1
4	Simplify $4m^6 \times 3m^2$. **A. $12m^8$** B. $7m^8$ C. $12m^{12}$ D. $12m^4$	1
5	What is the sum of the degrees of all the vertices in this network? Note: the degree of a vertex is the number of edges that meet at that vertex. A. 6 B. 8 C. 12 **D. 16** 	1

Short Answer:

	Question	Working Out	Mark
6	Expand and simplify: $3(4x - 5) - 2(x - 6)$	$= 12x - 15 - 2x + 12$ **$10x - 3$**	2

Marking guide: answers in red

Section 3: Mixed: Chapters 1-8 (continued)

	Question	Working Out	Mark
7	Liam invests \$7250 for 3 years at 4.2% p.a. simple interest. Calculate the interest he earns. Note: Simple interest: $I = Prn$	$I = \$7250 \times 0.042 \times 3$ **\$913.50**	2
8	Find the value of x, correct to 2 decimal places.  Note: $\sin\theta = \frac{\text{opp}}{\text{hyp}}$, $\cos\theta = \frac{\text{adj}}{\text{hyp}}$, $\tan\theta = \frac{\text{opp}}{\text{adj}}$	$\sin 38° = \frac{x}{14.5}$ $x = 14.5 \times \sin 38° = 8.9271\ldots$ **8.93 cm**	2
9	a. Write 0.000 724 in scientific notation. b. Evaluate $(4.8 \times 10^6) \times (3.5 \times 10^{-2})$, giving your answer in scientific notation.	a. Move the decimal point 4 places to the right. **7.24×10^{-4}** b. $= 4.8 \times 3.5 \times 10^6 \times 10^{-2}$ $= 16.8 \times 10^4$ **1.68×10^5**	1 1
10	Solve each equation. a. $5x - 8 = 27$ b. $7m - 2(m + 3) = 3m + 10$	a. $5x = 35$ ***x* = 7** b. $7m - 2m - 6 = 3m + 10$ $5m - 6 = 3m + 10$ $2m = 16$ ***m* = 8**	1 1

Marking guide: answers in red

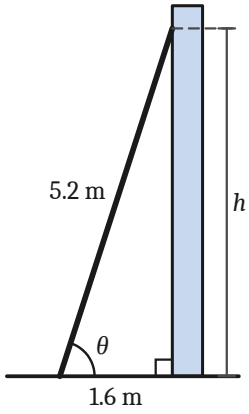
Section 3: Mixed: Chapters 1-8 (continued)

Worded Problems:

	Question	Working Out	Mark
11	Mia works in a bakery. Her normal rate of pay is $26.80 per hour, and she is paid according to the award shown below. Last week Mia worked 44 hours. **Pay rate** / **Hours worked** Normal rate / For 0 to 38 hours Time-and-a-half / For the next 3 hours Double time / For each hour after that		
a	Calculate Mia's pay for the first 38 hours.	38 × $26.80 **$1018.40**	1
b	Calculate Mia's total pay for the 44 hours.	Time-and-a-half: 3 × 1.5 × $26.80 = $120.60 Double time: 3 × 2 × $26.80 = $160.80 Total: $1018.40 + $120.60 + $160.80 **$1299.80**	2

Marking guide: answers in red

Section 3: Mixed: Chapters 1-8 (continued)

	Question	Working Out	Mark
12	A ladder 5.2 m long leans against a vertical wall. The foot of the ladder is 1.6 m from the wall, on level ground. Note: Pythagoras' theorem: $c^2 = a^2 + b^2$ Note: $\sin\theta = \frac{\text{opp}}{\text{hyp}}$, $\cos\theta = \frac{\text{adj}}{\text{hyp}}$, $\tan\theta = \frac{\text{opp}}{\text{adj}}$ 		
a	Find h, the height the ladder reaches up the wall, correct to 2 decimal places.	$h^2 = 5.2^2 - 1.6^2 = 27.04 - 2.56 = 24.48$ $h = \sqrt{24.48} = 4.9477\ldots$ **4.95 m**	1
b	Find the angle θ that the ladder makes with the ground, correct to the nearest degree.	$\cos\theta = \frac{1.6}{5.2}$ $\theta = \cos^{-1}(1.6 \div 5.2) = 72.08\ldots°$ **72°**	1
c	For safety, a ladder should make an angle of 75° with the ground. A longer ladder is placed at 75° so that its top reaches 6.5 m up the wall. How long is this ladder? Answer correct to 2 decimal places.	$\sin 75° = \frac{6.5}{L}$ $L = 6.5 \div \sin 75° = 6.7293\ldots$ **6.73 m**	1
	Total		**/21**